

EP02 — What is an algorithm?

HINGLISH (Roman)

01 Title

Algorithm **kya hota hai**? Sunne mein technical aur scary lagta hai — par ek secret bataun? Aap har roz algorithms use karte ho, bina jaane. Chalo ise simple karte hain.

[CUE] Title pe ~2s ruko. Friendly, confident — "ye lagta hai usse aasaan hai".

02 The Hook

Aaj 2-minute Maggi banayi? Toh aapne ek algorithm chalaya. Paani ubaalo, masala daalo, Maggi daalo, do minute ruko. **Steps, ek fixed order mein**, ek result ke liye. Algorithm **bilkul** yahi hai — ek recipe.

[CUE] Energy high. Steps ko ungliyon pe count karo jab bolo.

03 Aaj ka sawaal

Toh aaj ka sawaal: algorithm **exactly** hai kya — aur computer science ise itna important kyun maanti hai? Chalo pata karte hain.

[CUE] Thoda slow. Sawaal ko ek beat ke liye baithne do.

04 Idea 1 — Precise recipe

Idea ek. Algorithm bas ek **step-by-step recipe** hai, ek specific problem solve karne ke liye. Chai banana dekho: paani ubaalo, chai-cheeni daalo, doodh daalo, do minute ubaalo, chhaan lo. Ek clear set of steps, order mein, jo kaam pura karte hain. Algorithm bas itna hai.

[CUE] Recipe block ki taraf point karo; steps rhythm mein padho.

05 Idea 2 — Clear, ordered, finite

Idea do — par steps ke teen rule hote hain. Wo **clear** hone chahiye: "thoda sa daalo" nahi — kitna, exactly? Wo **ordered** hone chahiye: paani ubaalne se pehle Maggi daal di, toh disaster — order matters. Aur wo **finite** hone chahiye: recipe khatam honi chahiye. Computer aapke steps **literally** follow karta hai, toh vague hue toh wo break ho jaata hai.

[CUE] clear / ordered / finite — teen alag beats pe zor do.

06 Idea 3 — Efficiency

Idea teen — aur isiliye CS algorithms pe obsessed hai. **Same task** fast bhi ho sakta hai, slow bhi. Maan lo 1000-page dictionary mein ek word dhoondhna hai. Page-by-page jao, toh hazaar tak checks. Par beech wala page kholo, dekho word pehle hai ya baad mein, aur jump karo — sirf das checks. Dono sahi hain. Ek **sau guna tez** hai.

[CUE] Do bars (1000 vs ~10) reveal karo jab compare karo.

07 Idea 4 — Queue (FIFO)

Idea chaar — computers **kaam** ko bhi simple rules se organise karte hain. Samose ki line socho: jo pehle aata hai, use pehle milta hai. Yahi ek **queue** hai — CS mein hum ise **FIFO** kehte hain, First In First Out. Print jobs, ticket counter, yahaan tak ki aapke WhatsApp messages — sab queue se handle hote hain.

[CUE] Samose ki line left-to-right trace karo: "in... in... in... out".

08 Why it matters

Aur ye har jagah hai. Google Maps ek algorithm se aapka shortest route nikaalta hai. YouTube ek algorithm se aapka agla video chunta hai. UPI ek algorithm se do second mein payment verify karta hai. Sab kuch — bas clever step-by-step algorithms, jo ek second mein millions baar chalte hain.

[CUE] Teen examples tezi se; ye familiar hain, land hone do.

09 Common myth

Ab myth. Log sochte hain algorithm ek scary maths hai jo sirf coders aur AI use karte hain. Galat. Algorithm bas **clear steps** hain kisi kaam ke liye — aur aap inhe roz use karte ho. Code likhna sirf un steps ko itna **precise** likhna hai ki computer follow kar sake.

[CUE] Myth pe ek beat ke baad green "Reality" card reveal karo.

10 Recap

Quick recap. Ek: algorithm = **precise, ordered, finite steps** kisi problem ko solve karne ke liye. Do: same task ka fast ya slow algorithm ho sakta hai — aur fast wala chunna hi computer science ka dil hai.

[CUE] Crisp. Ye do lines yaad rakhni hain.

11 CTA

Toh agli baar koi "algorithm" bole, bas socho: recipe. Simple. Agar ye click hua, toh Katixo KhojLab ko subscribe karo. Agla sawaal: internet actually aapke phone tak pahunchta kaise hai? Milte hain agle video mein.

[CUE] Warm, friendly. Subscribe line pe awaaz mein smile.
